# Optimizing Hybrid RIS-Aided ISAC Systems in V2X Networks: A Deep Reinforcement Learning Method for Anti-Eavesdropping Techniques

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***Abstract*—Physical layer security (PLS) technique is expected to play a crucial part in the vehicle-to-everything (V2X) networks, by offering secure transmission to protect confidential information from potential eavesdropper. Considering a hybrid active-passive reconfigurable intelligent surfaces (RISs)-enhanced integrated sensing and communication (ISAC) system, this paper proposes a novel secure scheme for transmitting confidential information and performing radar sensing, where vehicle-to-vehicle (V2V) links share the spectrum resource preoccupied by vehicle-to-infrastructure (V2I) links. We aim to optimize the sum secrecy rate of V2I links by jointly designing the transmit beamforming of RSU, the radio spectrum reuse scheme of V2X links, and active and passive reflection beamforming of hybrid RIS. With above optimization, the proposed approach can enhance secure communication performance of V2I links while guaranteeing the communication quality of V2V links and target sensing capacity of RSU. Since the system model is dynamic, and it is difficult to handle the nonconvex problem, an efficient hierarchical twin delayed deep deterministic policy gradient (HTD3) method is developed to learn the secure beamforming and spectrum sharing strategies against potential eavesdropping. The proposed method decomposes the spectrum allocation into the deep Q-network procedure and designs the secure beamforming variables by employing the TD3 algorithm. Numerical results exhibit that given a sufficient power budget of hybrid RIS, our HTD3-based method enhances both the secure communication performance of V2I links and radar detection capability of RSU compared with the existing learning methods.**

***Index Terms*—Vehicular network, deep reinforcement learning, integrated sensing and communication (ISAC), secure communication, hybrid active-passive reconfigurable intelligent surface.**

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## I. Introduction

DRIVEN by the rapid advancement of wireless communication techniques, vehicle-to-everything (V2X) networks, are new scheme to provide leading edge applications, for example vehicular entertainment and intelligent driving [1], [2]. To implement these applications, intelligent vehicles are required to have communication and sensing capabilities. Therefore, it is important to deploy integrated sensing and communication (ISAC) system in mobile V2X networks for high rate communication and high-precision positioning [3]. ISAC systems have the ability to realize multi-view sensing information sharing among intelligent vehicles and roadside units (RSUs), enhancing the security and efficiency of mobile vehicular communication networks [4]. Since the ISAC waveforms also carry private information for the secure users, the radar target, as a malicious wiretapper, for example, an illegitimate user, might intercept the communication data intended for secure users [5]. In the V2X networks, confidentiality of communication data must be ensured since the disclosure of user information may result in severe consequences, for example illegal tracking, property damage, and traffic accidents [6]. For this purpose, physical layer security (PLS) techniques are needed for the dual functional operation in security-critical V2X communication system designs.

To address the potential security problem in ISAC systems, in [7], the multiple input multiple output (MIMO) system was devised to convey a mixture of two different waveforms, containing communication information for the authorized receivers and artificial noise (AN) to disturb the eavesdropper (Eve), both of which can be employed for radar target detection. In [8], the constructive interference (CI) method was used to optimize the dual-function waveform for utilizing the intrinsic multiuser interference (MUI) as beneficial power to secure ISAC systems. The AN and CI methods were jointly used to optimize secure performance under the hypothesis of imperfect channel state information (CSI) [9], which was shown to outperform the traditional AN-assisted security design. The presence of malicious

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eavesdroppers is considered in ISAC-based V2X network, and a power control scheme was proposed to guarantee secure communication [10]. To improve the vehicular communication performance against smart eavesdropper, a novel resource allocation strategy was developed to improve the system secrecy rate while ensuring radar sensing requirements [11].

Recently, reconfigurable intelligent surface (RIS) [12], as a promising technique, has been developed to enhance the spectrum efficiency and satisfy the PSL of V2X communications. RIS is capable of reducing high path loss in high-frequency bands to improve vehicular network connectivity as well as obtain performance gains by providing programmable propagation environment [13], [14]. The deployment of RIS in enhancing the quality of service (QoS) of vehicle-to-infrastructure (V2I) communications was confirmed in [15]. The work of [16] verified the improvement of security and reliability in V2X communications with the help of the RIS. The effectiveness of employing RIS-aided unmanned aerial vehicle (UAV) in V2X networks was presented in [17]. To optimize the performance of simultaneous transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) in vehicular communication networks, a novel scheme is to deploy STAR-RIS on the surface of intelligent vehicle to enhance sensing and communication capabilities [18]. ISAC systems have the problem of information leakage when employing the dual-function waveforms to sense the environment. RIS-assisted beamforming design methods have been proposed for PLS improvement in the ISAC systems [19], [20], [21], [22]. The secure performance was optimized by jointly devising the beamforming at the base station (BS) and the reflection weights at the active RIS, subject to the radar detection performance requirement as well as the power budget constraints at the BS and active RIS [19]. Assuming that an Eve possibly occurs in a specific location and the eavesdropping channel is imperfect known, an optimization problem was established to optimize the system secrecy rate of the authorized user [20]. The authors studied a RIS-assisted ISAC secure system where BS concurrently executes multi-user multi-input single-output (MU-MISO) secure transmission and senses a potential eavesdropping target [21]. By jointly devising the transmit beamforming, the artificial noise (AN) waveforms and the passive RIS weights, secure transmission performance was optimized in [22]. Deep reinforcement learning (DRL) procedure was used to obtain the optimal learning scheme.

Owing to the passive reflection characteristics of the RIS, the employment of RIS would cause the multiplicative path loss effect, thus resulting in the serious communication and sensing performance deterioration. The fully active RIS is capable of utilizing amplifiers to tackle the serious channel attenuation. However, it requires extra power consumption to improve the strength of the reflected waveforms and noise at the active RIS [23]. To overcome the aforementioned issues, hybrid active-passive RIS including a certain number of active units was proposed [24], [25], which serves the dual purpose of reflecting and amplifying incoming signals and achieves a significant improvement of system performance. The robust beamforming optimization for a hybrid RIS-enabled ISAC system was proposed in [24]. The work of [25] developed a design scheme to optimize the date rate for hybrid RIS-assisted communication system that operates in non-line-of sight (NLoS) channels. The jointly design of the hybrid precoders and the hybrid RIS weights for optimizing the weighted sum data rate, subject to the overall system power constraint, was proposed in [26]. The authors of [15] also considered adding active reflection units to passive RIS and then employed the hybrid architecture for energy transmission, and verified that the hybrid RIS can provide superior performance than fully passive RIS.

Compared with the traditional optimization algorithms, the artificial intelligence (AI) approaches have revealed superior robustness to handle complex problems with inaccurate modeling, multiuser couplings and uncertain dynamics [27], [28], [29], [30]. In [31], the deep Q-learning (DQN) technique was employed to realize secure beamforming in the presence of potential eavesdropper. DQN belongs to a deep learning scheme that calculates the system state and is developed to handle these discrete action problems. Furthermore, the authors of [32] and [27] developed DRL method to handle the continuous control problem, and the reflecting beamforming was designed efficiently. In [33], a DRL-based technique was proposed to allocate the spectrum resource for the relay-aided cellular V2X communications. Different from centralized training strategy, multi-agent DRL approach is capable of compromising cooperative and competitive trade-offs of agents to realize a dynamic balance in mobile vehicular networks [34]. Since the consideration of intelligent resource allocation makes the RIS-aided orthogonal frequency division multiplexing (OFDM) system more complex, a twin delayed deep deterministic policy gradient (TD3)-based technique [35], [36] was developed to enhance learning efficiency. The work of [19] proposed a DRL-based strategy to handle the joint optimization problem of radar sensing and communication in the STAR-RIS-enhanced vehicular networks. However, these studies [19], [27], [29], [30], [33], [34], [36] only considered to optimize the achievable transmission capacity without taking into account the secure communication and outdated CSI condition in the system models. To the best of our knowledge, PLS optimization technique based on DRL method for hybrid RIS-enhanced secure V2X networks has not been studied before.

Motivated by the above discussions, we study the PLS of hybrid RIS-enhanced ISAC secure system in realistic time-varying V2X networks. In particular, assuming that the malicious target might intercept the private information intended to the cellular vehicle users (CVUs), a novel hierarchical DRL-based technique is proposed to design an efficient scheme for securing V2X networks under the condition of outdated CSI. An inspection into two representative DRL techniques, for instance, DQN and TD3, displays that they depend on the double Q-network framework. The system architecture brings tight coupling between double Q-networks and thus causes performance loss. The phenomenon motivates us to propose a hierarchical TD3 (called HTD3) procedure to enhance the learning efficiency. The main idea of the HTD3 is to minimize the action space by introducing a hierarchical framework, namely, the discrete variables are designed by the DQN procedure while the continuous variables are optimized by TD3 algorithm. Our main contributions of this paper are summarized as follows.

- We are the first to adopt hybrid RIS-enhanced ISAC into secure vehicular communication networks with high

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mobility, where multiple V2V links reuse the radio spectrum already occupied by V2I links. In addition, the outage probability is employed to guarantee the reliability of hybrid RIS-enhanced V2V links.

- We investigate the sum V2I secrecy rate maximization problem, where the transmit beamforming of RSU, the spectrum reuse scheme of V2X links, and active and passive reflection weights of hybrid RIS are jointly designed while ensuring the V2V communication quality requirement and desired RSU target sensing capability under a particular resource constraint.
- Since the system model is highly dynamic, an efficient HTD3 learning algorithm is proposed to obtain the optimal anti-eavesdropping strategy, where the outer-loop DQN procedure can quickly obtain the optimal spectrum allocation, the inner-loop TD3 procedure has the ability to optimize the secure beamforming variables.
- Simulation results are offered to validate that the hybrid RIS-enhanced V2X network improves the anti-eavesdropping communication performance compared to conventional RIS-based systems. Moreover, the achieved numerical results confirm that the employment of hybrid RIS in time-varying vehicular networks can compensate for the channel losses owing to the mobility of connected vehicles. Furthermore, the convergence and efficiency of the proposed method are also demonstrated.

The rest of this article is organized as follows: Section II introduces the anti-eavesdropping model in hybrid RIS-enhanced vehicular networks. In Section III, the joint beamforming optimization and spectrum allocation problem is proposed, and the HTD3 learning procedure is designed to handle the formulated DRL problem. Section IV offers simulation experiments to confirm the advantage of the devised HTD3 learning procedure. Finally, some conclusions are presented in Section V.

*Notation:* Lowercase letters denote scalars. Boldface uppercase and lowercase letters signify matrices and vectors, respectively. The symbols $(\cdot)^*$, $(\cdot)^T$ and $(\cdot)^H$ signify the conjugate, transpose and the conjugate transpose operators, respectively. $\|\mathbf{s}\|$ is the Euclidean norm of $\mathbf{s}$. $\mathbf{I}$ and $\mathbf{0}$ denote a matrix with proper size which all the elements are equal to one and zero respectively. $\mathbb{E}$ signifies the expectation operator. $\nabla(\cdot)$ signifies the first-order partial derivative. The distribution of a complex Gaussian with zero mean and variance $\sigma^2$ is represented by $\mathcal{CN}(0,\sigma^2)$.

## II. System Model

We consider a hybrid RIS-enhanced vehicular network with V2I and V2V links that offers support for intelligent driving services, such as security information sharing and environmental perception, as described in Fig. 1. In particular, the studied V2X network consists of $M$ single antenna cellular-vehicle users (CVUs) that communicate with the multi-antenna ISAC-based RSU using V2I links, and $L$ pairs of device-to-device vehicle users (DVUs) that connect with each other using V2V links. Furthermore, a malicious radar target acts as an Eve, and utilizes single antenna for intercepting the confidential information

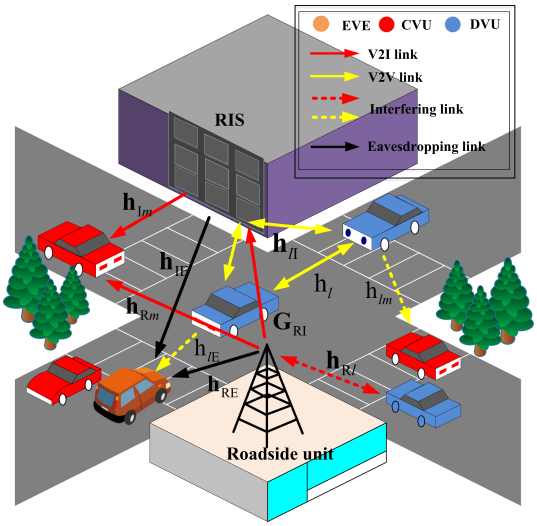


Fig. 1. A hybrid RIS-enhanced secure V2X network based on ISAC framework.

transmission between the RSU and CVUs. The RSU equipped with $N_t$ transmitting and $N_r$ receiving antennas, synchronously serves $M$ CVUs and detects an Eve by emitting an ISAC probing waveform.

The hybrid RIS matrix[1] is given by $\mathbf{\Theta} = \text{diag}\{\theta_1, \ldots, \theta_{N_\text{I}}\}$, and considering $\mathbb{Q}$ as the set of active units in the hybrid RIS, $\theta_n$ can be denoted as

$$\theta_n = \begin{cases} |a_n|\, e^{j\mu_n}, n \in \mathbb{Q}, \\ e^{j\mu_n}, n \notin \mathbb{Q}, \end{cases} \tag{1}$$

where $a_n$ denotes the amplitude of the signal generated by active unit, $\mu_n$ is the phase of the signal. The hybrid RIS matrix can be viewed as the sum of two matrices, that is, $\mathbf{\Theta} = \mathbf{\Phi} + \mathbf{\Psi}$, in which $\mathbf{\Phi} = \text{diag}\{\phi_1, \ldots, \phi_{N_\text{I}}\}$ and $\mathbf{\Psi} = \text{diag}\{\varphi_1, \ldots, \varphi_{N_\text{I}}\}$ are the matrices with respect to passive units and active units, respectively. $\phi_n$ and $\varphi_n$ can be represented as

$$\phi_n = \begin{cases} 0, n \in \mathbb{Q}, \\ e^{j\mu_n}, n \notin \mathbb{Q}, \end{cases} \tag{2}$$

and

$$\varphi_n = \begin{cases} |a_n|\, e^{j\mu_n}, n \in \mathbb{Q}, \\ 0, n \notin \mathbb{Q}. \end{cases} \tag{3}$$

### A. Channel Model

We consider Mode 4 described in the cellular vehicular network framework, where connected vehicle users have a spectrum resource pool that they can choose V2V transmission independently [37]. The channel response from the RSU to the $m$th CVU is represented by $\mathbf{h}_{\text{R}m} = \sqrt{L_0 d_{\text{R}m}^{-\alpha_{\text{R}m}}}\,\bar{\mathbf{h}}_{\text{R}m} \in \mathbb{C}^{N_t \times 1}$, where $L_0$ represents the path loss at the reference distance of 1 m. $d_{\text{R}m}$ and $\alpha_{\text{R}m}$ are the distance and the path loss factor of RSU-CVU$_m$ link, respectively. Each element of $\bar{\mathbf{h}}_{\text{R}m}$ obeys a zero mean Gaussian distribution. The channel response between the RSU

[1]The hybrid RIS includes $N_h$ horizontally and $N_v$ vertically reflection units. Hence the total number of reflection units is $N_\text{I} = N_h \times N_v$.

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and the hybrid RIS obeys a Rician distribution [15], which is signified by

$$\mathbf{G}_{\mathrm{RI}} = \sqrt{L_0 d_{\mathrm{RI}}^{-\alpha_{\mathrm{RI}}}}\left(\sqrt{\frac{\kappa_{\mathrm{RI}}}{1+\kappa_{\mathrm{RI}}}}\mathbf{G}_{\mathrm{RI}}^{\mathrm{LoS}} + \sqrt{\frac{1}{1+\kappa_{\mathrm{RI}}}}\mathbf{G}_{\mathrm{RI}}^{\mathrm{NLoS}}\right), \tag{4}$$

where $d_{\mathrm{RI}}$ and $\alpha_{\mathrm{RI}}$ indicate the distance and the path loss factor of the RSU-RIS link, respectively. $\kappa_{\mathrm{RI}}$ signifies the corresponding Rician factor. The line-of-sight (LoS) part $\mathbf{G}_{\mathrm{RI}}^{\mathrm{LoS}} \in \mathbb{C}^{N_I \times N_t}$ is composed of the uniform linear array (ULA) array response. All elements of the NLoS part $\mathbf{G}_{\mathrm{RI}}^{\mathrm{NLoS}}$ obey a zero mean Gaussian distribution. The channel response from the hybrid RIS to the $m$th CVU is given by $\mathbf{h}_{\mathrm{I}m} \in \mathbb{C}^{N_I \times 1}$, which obeys a Rician distribution analogous to Eq. (4). Concerning the LoS part of the RIS-CVU channel, the angle of departure (AoD) at the hybrid RIS relies primarily on the position of the CVU, namely, $\vartheta^{\mathrm{AoD}} = \arccos(|L_m - L_{\mathrm{R}}|/d_{\mathrm{R}m})$ [38], where $L_m$ and $L_{\mathrm{R}}$ signify the locations of the $m$th CVU and the hybrid RIS, respectively. Analogously, the channel response of the $l$th V2V link, the channel response between the $l$th V2V transmitter and the hybrid RIS, which are defined by $h_l$ and $\mathbf{h}_{l\mathrm{I}} \in \mathbb{C}^{N_I \times 1}$, respectively.

Furthermore, let $\mathbf{h}_{\mathrm{R}l} \in \mathbb{C}^{N_t \times 1}$ signify the interfering channel power gain from the RSU to the $l$th V2V link, let $h_{lm}$ signify the interfering channel response from the $l$th V2V link to the $m$th CVU. The channel response from the RSU to the Eve $\mathbf{h}_{\mathrm{RE}} \in \mathbb{C}^{N_t \times 1}$, the interfering channel response between the $l$th V2V transmitter and the Eve $h_{l\mathrm{E}}$, the channel response from the hybrid RIS to the Eve $\mathbf{h}_{\mathrm{IE}} \in \mathbb{C}^{N_I \times 1}$, are similarly defined.

In the considered vehicular network, it is difficult for RSU and hybrid RIS to gain perfect CSI owing to the mobility of the vehicle and the existence of transmission and processing delay [39], [40]. As a result, CSI may be outdated when RSU and hybrid RIS convey information to CVUs. If the outdated CSI is used for secure beamforming, it would result in an adverse impact on the demodulation at the CVUs, thus generating significant performance degradation [40]. Hence, it is appropriate to analyze outdated CSI in the hybrid RIS-enhanced V2X network.[2] Let $T_{\mathrm{delay}}$ signify the time delay between the outdated CSI and the actual CSI. Therefore, the relationship between the outdated channel power gain $\mathbf{h}(t)$ and the actual channel power gain $\mathbf{h}(t + T_{\mathrm{delay}})$ is given by

$$\mathbf{h}\left(t + T_{\mathrm{delay}}\right) = \rho_d \mathbf{h}(t) + \sqrt{1-\rho_d^2}\hat{\mathbf{h}}\left(t + T_{\mathrm{delay}}\right). \tag{5}$$

All elements of $\hat{\mathbf{h}}(t + T_{\mathrm{delay}})$ obey a zero mean Gaussian distribution. $\rho_d (0 < \rho_d \leq 1)$ denotes the outdated autocorrelation coefficient, which is expressed by $\rho_d = J_0(2\pi f_D T_{\mathrm{delay}})$, with $J_0(\cdot)$ and $f_D$ being the zeroth-order Bessel function and the Doppler spread, respectively. Notice that $\rho_d = 1$ suggests the delay effect does not exist, whereas $\rho_d = 0$ signifies no CSI.

[2]It is worth noting that the outdated CSI of all channels can be achieved by existing channel estimation methods. It is appropriate to suppose that the outdated CSI of Eve is known in the RIS-enhanced V2X network when the Eve is an active user but untrusted by all authorized users [41]. Under these assumptions, our analysis provides information on the best secure communication performance upper bounds for more complex scenarios.

### B. Communications Model

We begin with the signals emitted from the RSU. The dual function transmit beam signal at the RSU in time step $n$ is denoted as

$$\mathbf{x}(n) = \sum_{m \in M} \mathbf{v}_m s_m(n) + \mathbf{w}_r c(n), \tag{6}$$

where $s_m(n)$ and $c(n)$ signify the communication symbol intended to the $m$th CVU and the individual radar waveform with unit power, respectively. $\mathbf{v}_m \in \mathbb{C}^{N_t \times 1}$ and $\mathbf{w}_r \in \mathbb{C}^{N_t \times 1}$ indicate the communication beamforming for the $m$th CVU and the adjustable radar beamforming vector, respectively. Without loss of generality, we consider the communication waveform $\mathbf{s}(n) = [s_1(n), \ldots, s_M(n)]^T$ is a wide-sense stationary random process with zero-mean and unit power, and uncorrelated with each other, that is, $\mathbb{E}[\mathbf{s}(n)\mathbf{s}^H(n)] = \mathbf{I}$. Besides, the communication symbols are uncorrelated with the radar waveform, i.e, $\mathbb{E}[\mathbf{s}(n)c(n)] = \mathbf{0}$.

Active V2V links reuse the spectrum resource of the V2I links, hence mutual interference should be properly considered in communication model. In this case, when the RSU sends sensitive information to the $m$th CVU, the $m$th CVU receives the signal transmitted from its connected RSU, the reflected signal from the hybrid RIS and the interference from the V2V transmitter occupied by the $m$th V2I link, which is expressed as

$$\begin{aligned} y_m^c(n) = &\underbrace{(\mathbf{h}_{\mathrm{I}m}^H \mathbf{\Theta} \mathbf{G}_{\mathrm{RI}} + \mathbf{h}_{\mathrm{R}m}^H)\mathbf{v}_m s_m(n)}_{\text{desired signal}} \\ &+ \underbrace{\sum_{l=1}^{L} \rho_{ml}(\mathbf{h}_{\mathrm{I}m}^H \mathbf{\Theta} \mathbf{h}_{l\mathrm{I}} + h_{lm}) s_l(n)}_{\text{V2V interference signal}} + n_m^c(n), \end{aligned} \tag{7}$$

where $s_l(n)$ represents the unit power information symbol of the $l$th V2V transmitter, with $\mathbb{E}\{|s_l(n)|^2\} = 1$. $n_m^c(n) \sim \mathcal{CN}(0, \sigma_{c,m}^2), \forall m \in M$ is the additive Gaussian noise at the $m$th CVU. $\rho_{ml}$ denotes the spectrum reuse factor with $\rho_{ml} = 1$ indicating the $l$th active V2V link shares the $m$th V2I link. When spectrum reuse is not executed, $\rho_{ml} = 0$. Concerning the $l$th V2V receiver, the received signal can be denoted as

$$\begin{aligned} y_l^c(n) = &\underbrace{(\mathbf{h}_{\mathrm{I}l}^H \mathbf{\Theta} \mathbf{h}_{l\mathrm{I}} + h_l) s_l(n)}_{\text{desired signal}} \\ &+ \underbrace{\sum_{m=1}^{M} \rho_{ml}(\mathbf{h}_{\mathrm{I}l}^H \mathbf{\Theta} \mathbf{G}_{\mathrm{RI}} + \mathbf{h}_{\mathrm{R}l}^H)\mathbf{v}_m s_m(n)}_{\text{V2I interference signal}} + n_l^c(n), \end{aligned} \tag{8}$$

where $n_l^c(n) \sim \mathcal{CN}(0, \sigma_{c,l}^2), \forall l \in L$. When the Eve attempts to intercept the private information intended for the $m$th CVU, the received signal at the Eve can be represented as

$$y_m^E(n) = \underbrace{(\mathbf{h}_{\mathrm{IE}}^H \mathbf{\Theta} \mathbf{G}_{\mathrm{RI}} + \mathbf{h}_{\mathrm{RE}}^H)\mathbf{v}_m s_m(n)}_{\text{desired signal}}$$

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$$+\underbrace{\sum_{l=1}^{L}\rho_{ml}(\mathbf{h}_{\mathrm{IE}}^{H}\boldsymbol{\Theta}\mathbf{h}_{l\mathrm{I}}+\mathrm{h}_{l\mathrm{E}})s_l(n)}_{\text{V2V interference signal}}+n_m^{E}(n), \tag{9}$$

where $n_m^{\mathrm{E}}(n)\sim\mathcal{CN}(0,\sigma_{\mathrm{E},m}^2)$. Under the assumption of outdated CSI, the downlink signal-to-interference-plus-noise (SINR) of the $m$th CVU can be written as

$$\mathrm{SINR}_m(\mathbf{v},\boldsymbol{\Theta},\rho)=\frac{|(\mathbf{h}_{\mathrm{I}m}^{H}\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}+\mathbf{h}_{\mathrm{R}m}^{H})\mathbf{v}_m|^2}{|\sum_{l=1}^{L}\rho_{ml}(\mathbf{h}_{\mathrm{I}m}^{H}\boldsymbol{\Theta}\mathbf{h}_{l\mathrm{I}}+h_{lm})|^2+\sigma_{c,m}^2}. \tag{10}$$

The SINR of the $l$th V2V receiver is

$$\mathrm{SINR}_l(\mathbf{v},\boldsymbol{\Theta},\rho)=\frac{|(\mathbf{h}_{\mathrm{I}l}^{H}\boldsymbol{\Theta}\mathbf{h}_{l\mathrm{I}}+h_l)|^2}{|\sum_{m=1}^{M}\rho_{ml}(\mathbf{h}_{\mathrm{I}l}^{H}\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}+\mathbf{h}_{\mathrm{R}l}^{H})\mathbf{v}_m|^2+\sigma_{c,l}^2}. \tag{11}$$

Since the Eve is capable of intercepting any of $M$ CVUs' received signal, the received SINR of the Eve on the $m$th sub-band is

$$\mathrm{SINR}_m^{\mathrm{E}}(\mathbf{v},\boldsymbol{\Theta},\rho)=\frac{|(\mathbf{h}_{\mathrm{IE}}^{H}\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}+\mathbf{h}_{\mathrm{RE}}^{H})\mathbf{v}_m|^2}{|\sum_{l=1}^{L}\rho_{ml}(\mathbf{h}_{\mathrm{IE}}^{H}\boldsymbol{\Theta}\mathbf{h}_{l\mathrm{I}}+h_{l\mathrm{E}})|^2+\sigma_{\mathrm{E},m}^2}. \tag{12}$$

Consequently, the downlink data rate of the $m$th CVU is $R_m=\log_2(1+\mathrm{SINR}_m)$ and the data rate of the $l$th V2V link is $R_l=\log_2(1+\mathrm{SINR}_l)$. In addition, the data rate at the Eve on the $m$th frequency band is given by $R_{\mathrm{E},m}=\log_2(1+\mathrm{SINR}_m^{\mathrm{E}})$. The secure communication of the V2I link is the key performance criterion of interest. Therefore, the system secrecy rate from the RSU to the $m$th CVU can be defined as

$$R_{\mathrm{S},m}=\max\{0,R_m-R_{\mathrm{E},m}\}. \tag{13}$$

### C. Radar Model

As described in Fig. 13, the dual function waveform illuminates the radar target (i.e., Eve) through a direct path and reflected paths, and be backscattered to the RSU via all the transmission channels. The target echo is achieved by transmitting the dual function waveform via four various channels, i.e., RSU-Eve-RSU, RSU-RIS-Eve-RSU, RSU-Eve-RIS-RSU and RSU-RIS-Eve-RIS-RSU. Hence, the radar signal received at the RSU, which is composed of the target echo, the V2V interference signal and the additive noise, given by (14) shown at the bottom of this page, where $\mathbf{H}_a$, $\mathbf{H}_b$, $\mathbf{H}_{ab}$ and $\mathbf{H}_{ba}$ denote the target response matrices. They can be modeled as LOS and depend on azimuth angles [42], respectively, given by

$$\mathbf{H}_a=\mathbf{a}_r(\theta_a)\mathbf{a}_t^H(\theta_a),\mathbf{H}_b=\mathbf{b}(\theta_b)\mathbf{b}^H(\theta_b),$$
$$\mathbf{H}_{ab}=\mathbf{a}_r(\theta_a)\mathbf{b}^H(\theta_b),\mathbf{H}_{ba}=\mathbf{b}(\theta_b)\mathbf{a}_t^H(\theta_a), \tag{15}$$

where $\mathbf{a}_t(\theta_a)$ and $\mathbf{a}_r(\theta_a)$ represent the transmit and receive steering vectors of the ISAC system, and $\mathbf{b}(\theta_b)$ signifies the reflection steering vector of the hybrid RIS. $\theta_a$ and $\theta_b$ denote the angles of the radar target (i.e., Eve) relative to the RSU and the hybrid RIS, respectively. $\alpha_{\mathrm{E}}$ signifies the complex coefficient that includes path loss and radar cross section (RCS) of the radar target with $\mathbb{E}\{|\alpha_{\mathrm{E}}|^2\}=\zeta_{\mathrm{E}}^2$. $\mathbf{n}_{\mathrm{R}}(n)\sim\mathcal{CN}(0,\sigma_R^2\mathbf{I}_{N_r})$ is the additive Gaussian noise at the RSU. Let $\mathbf{T}_{\mathrm{E}}=\mathbf{H}_a+\mathbf{H}_{ab}\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}+\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}\mathbf{H}_{ba}+\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}\mathbf{H}_b\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}$, the radar output SINR at the RSU is given by

$$\mathrm{SINR}_{\mathrm{E}}^{\mathrm{R}}=\frac{\zeta_{\mathrm{E}}^2|\mathbf{T}_{\mathrm{E}}\mathbf{w}_r|^2}{|\sum_{l=1}^{L}\rho_{ml}(\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}\mathbf{h}_{l\mathrm{I}}+\mathbf{h}_{l\mathrm{R}})|^2+\sigma_R^2}. \tag{16}$$

Accordingly, the radar mutual information (MI) can be written as

$$\mathrm{MI}=\log_2(1+\mathrm{SINR}_{\mathrm{E}}^{\mathrm{R}}). \tag{17}$$

It is worth noting that MI could effectively measure the parameter estimation performance of radar systems. More importantly, in contrast to other metrics, MI has a more brief structure, which is helpful to our analysis of the influence of uncertainty information on radar sensing performance and the design of algorithms [43]. The receive signal at the hybrid RIS is given by

$$\mathbf{y}_{\mathrm{RIS}}(n)=\mathbf{G}_{\mathrm{RI}}\left(\sum_{m\in M}\mathbf{v}_m s_m(n)+\mathbf{w}_r c(n)\right)+\sum_{l\in L}\mathbf{h}_{l\mathrm{I}}s_l(n)+\mathbf{n}_{\mathrm{I}}(n), \tag{18}$$

where $\mathbf{n}_{\mathrm{I}}(n)\sim\mathcal{CN}(0,\sigma_I^2\mathbf{I}_{N_I})$ is the additive Gaussian noise at the hybrid RIS. Hence, the desired power of the hybrid RIS is

$$P_{\mathrm{RIS}}=\mathrm{tr}\left(\boldsymbol{\Psi}\mathbf{y}_{\mathrm{RIS}}\mathbf{y}_{\mathrm{RIS}}^H\boldsymbol{\Psi}^H\right)=\mathrm{tr}\left(\boldsymbol{\Psi}\boldsymbol{\Gamma}\left(\{\mathbf{v}_m\}_{m=1}^{M},\mathbf{w}_r\right)\boldsymbol{\Psi}^H\right), \tag{19}$$

where

$$\boldsymbol{\Gamma}\left(\{\mathbf{v}_m\}_{m=1}^{M},\mathbf{w}_r\right)=\mathbf{G}_{\mathrm{RI}}\left(\sum_{m\in M}\mathbf{v}_m\mathbf{v}_m^H+\mathbf{w}_r\mathbf{w}_r^H\right)\mathbf{G}_{\mathrm{RI}}^H+\sum_{l\in L}\mathbf{h}_{l\mathrm{I}}\mathbf{h}_{l\mathrm{I}}^H+\sigma_I^2\mathbf{I}_{N_I}. \tag{20}$$

$$\mathbf{y}_{\mathrm{E}}^{R}(n)=\alpha_{\mathrm{E}}\left(\underbrace{\mathbf{H}_a}_{RSU-Eve-RSU}+\underbrace{\mathbf{H}_{ab}\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}}_{\text{RSU-RIS-Eve-RSU}}+\underbrace{\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}\mathbf{H}_{ba}}_{\text{RSU-Eve-RIS-RSU}}+\underbrace{\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}\mathbf{H}_b\boldsymbol{\Theta}\mathbf{G}_{\mathrm{RI}}}_{\text{RSU-RIS-Eve-RIS-RSU}}\right)\mathbf{w}_r c(n)$$
$$+\underbrace{\sum_{l=1}^{L}\rho_{ml}(\mathbf{G}_{\mathrm{RI}}^H\boldsymbol{\Theta}h_{l\mathrm{I}}+\mathbf{h}_{l\mathrm{R}})s_l(n)}_{\text{V2V interference signal}}+\mathbf{n}_R(n). \tag{14}$$

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### D. Problem Formulation

As explained in the preceding section, each V2I link is primarily utilized for the efficient transmission of security-critical information. Hence, we focus on optimizing the sum V2I secrecy rate, that is, $\sum_{m=1}^{M} R_m^{\mathrm{S}}$, while guaranteeing RSU sensing capability. Furthermore, the V2V transmission reliability is the key performance index of interest. The outage probability is employed to guarantee the reliability of V2V communications, given by

$$\mathrm{P}_r\left(R_l \leq R_{\min}^l\right) \leq P_{\max}^{\text{outage}}, \tag{21}$$

where $R_{\min}^l$ denotes the minimum data rate of the $l$th V2V link, $P_{\max}^{\text{outage}}$ signifies the maximum allowable outage probability. From this foundation, the secure vehicular communication problem includes the joint optimization of the communication and radar beamforming vectors $\{\mathbf{v}_m\}_{m=1}^M, \mathbf{w}_r$, the passive and active reflection matrices $\mathbf{\Phi}, \mathbf{\Psi}$, the spectrum allocation scheme for the V2X links $\boldsymbol{\rho} = \{\rho_{ml}, \forall m, l\}$. Hence, the problem that guarantees the desired transmission security and quality for hybrid RIS-enhanced vehicular networks while providing radar sensing capability, is cast as

$$\max_{\{\mathbf{v}_m\}_{m=1}^M, \mathbf{w}_r, \mathbf{\Phi}, \mathbf{\Psi}, \boldsymbol{\rho}} \sum_{m=1}^{M} R_{\mathrm{S},m} \tag{22a}$$

$$\text{s.t.} \sum_{m\in M} \|\mathbf{v}_m\|^2 + \|\mathbf{w}_r\|^2 \leq P_{\max}^{\mathrm{R}}, \tag{22b}$$

$$\sum_{m\in M} \rho_{ml} \leq 1, \sum_{l\in L} \rho_{ml} \leq 1, \rho_{m,l} \in \{0,1\}, \forall m, l, \tag{22c}$$

$$\mathrm{MI} \geq \mathrm{MI}_{\min}, \tag{22d}$$

$$P_{\mathrm{RIS}} \leq P_{\max}^{\mathrm{RIS}}, \tag{22e}$$

$$|\theta_n| = 1, \text{for } n \notin Q, \tag{22f}$$

$$R_m \geq R_{\min}^m, \forall m, \tag{22g}$$

$$(21), \tag{22h}$$

where $P_{\max}^{\mathrm{R}}$ and $P_{\max}^{\mathrm{RIS}}$ signify the maximum power budgets of the RSU and the hybrid RIS, respectively. $\mathrm{MI}_{\min}$ denotes the given radar MI threshold, $R_{\min}^m$ is the minimum data rate of the $m$th V2I link. To elaborate further, (22c) illustrates the spectrum allocation strategy for the V2X links, that is, the frequency spectrum of one CVU is shared with at most one V2V link, whereas one V2V link only shares the spectrum of at most one V2I link. The constraint in (22d) is considered to secure a particular radar detection criterion. (22e) and (22f) illustrate the constraints of active and passive RIS reflection units, respectively. (22g) is set to satisfy the worst-case data rate required for each V2I link.

## III. DRL Based Secure Beamforming and Spectrum Allocation

The resulting problem (22) is not easy to handle since it is non-convex and the variables are coupled with each other. Moreover, in hybrid RIS-enhanced secure V2X network, the channel conditions would vary dynamically. In addition, problem (22) belongs to single time slot optimization problem. Nevertheless, the conventional optimization procedures might converge to a non-optimal feasible solution since the historical state and the long term benefit are ignored. As a result, it is impracticable to exploit the conventional optimization procedures to acquire a valid secure communication strategy in dynamic vehicular communication network.

### A. Environmental Model

DRL is regarded as a dynamic programming tool which is employed to handle the sophisticated control problem by training the optimized strategy in complex systems [44]. On this basis, we recast the secure communication performance optimization problem as a Markov decision process (MDP) framework so that the model-free DRL procedure can be exploited. In DRL framework, the secure V2X network is considered as the external environment, the central control unit at the hybrid RIS-enhanced RSU is viewed as the learning agent. The system state, system action, and reward function for the secure vehicular communication problem are described as follows.

*System State:* Let $\mathcal{S}$ represent the state space. The state $s^{(t)} \in \mathcal{S}$ at the current time $t$ includes spectrum allocation, hybrid RIS weights, transmit beamforming vectors, radar MI, data rate of all CVUs and DVUs, and secrecy rate of all CVUs at the previous time $t-1$, as well as the outdated channel conditions corresponding to all $C$ RIS-enhanced V2X links at the current time $t$. Hence, the system state $s^{(t)}$ is given by

$$s^{(t)} = \left\{ \begin{array}{l} a^{(t-1)}, \left\{ (R_{\min}^m)^{(t-1)} \right\}_{m\in M}, \\ \left\{ (R_{\min}^l)^{(t-1)} \right\}_{l\in L}, \left\{ \mathbf{h}_c^{(t)} \right\}_{c\in \mathcal{C}} \end{array} \right\}, \tag{23}$$

where $a^{(t-1)}$ is the action at previous time $t-1$ including spectrum allocation, hybrid RIS weights, transmit beamforming vectors, radar MI, data rate of all CVUs and DVUs, and secrecy rate of all CVUs. $(R_{\min}^m)^{(t-1)}$ and $(R_{\min}^l)^{(t-1)}$ are the minimum date rates of the $m$th CVU and the $l$th DVU at time step $t-1$, respectively.

*System Action:* Let $\mathcal{A}$ represent the system action space. The learning agent selects the transmit beamforming vectors $\{\mathbf{v}_m\}_{m=1}^M$, $\mathbf{w}_r$ at the RSU, the passive and active reflection weights $\mathbf{\Phi}, \mathbf{\Psi}$ at the hybrid RIS, the coefficients of bandwidth resources allocated for V2X links $\boldsymbol{\rho}$. Therefore, the system action at time step $t$ is described as

$$a^{(t)} = \left\{ \left\{ \mathbf{v}_m^{(t)} \right\}_{m=1}^M, \mathbf{w}_r^{(t)}, \mathbf{\Phi}^{(t)}, \mathbf{\Psi}^{(t)}, \boldsymbol{\rho}^{(t)} \right\}. \tag{24}$$

Except the binary spectrum sharing variables $\boldsymbol{\rho}^{(t)}$, all other secure beamforming variables $\{\{\mathbf{v}_m^{(t)}\}_{m=1}^M, \mathbf{w}_r^{(t)}, \mathbf{\Phi}^{(t)}, \mathbf{\Psi}^{(t)}\}$ are continuous variables. The feasibility of the system action depends on the constraints of the hybrid RIS-enhanced secure vehicular communication optimization problem in (22).

*Reward function:* The reward function serves as an indicator to assess how well the secure vehicular communication policy is when the hybrid RIS-enhanced RSU performs a valid action. The secure communication performance is enhanced when the reward signal at training phase correlates with the predetermined

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goal. Hence, it is essential to devise a valid reward to enhance the vehicular communication security and quality. In our work, the reward function signifies the optimal object, and our aim is to optimize the sum secrecy rate of all CVUs while ensuring all legitimate vehicle users' communication quality requirement and RSU's target detection performance. Therefore, the security-critical reward function is written as

$$r = \underbrace{\sum_{m=1}^{M} R_{\mathrm{S},m}}_{\text{part1}} + \underbrace{\lambda_1 b^{\mathrm{RAD}}}_{\text{part2}} + \underbrace{\lambda_2 \sum_{m=1}^{M} b_m^{\mathrm{V2I}}}_{\text{part3}} + \underbrace{\lambda_3 \sum_{l=1}^{L} b_l^{\mathrm{V2V}}}_{\text{part4}}, \quad (25)$$

where

$$b^{\mathrm{RAD}} = \begin{cases} 1, \text{if } \mathrm{MI} - \mathrm{MI}_{\min} > 0, \\ 0, \text{otherwise}, \end{cases} \quad (26)$$

and

$$b_m^{\mathrm{V2I}} = \begin{cases} 1, \text{if } R_m - R_{\min}^m > 0, \forall m, \\ 0, \text{othersize}, \end{cases} \quad (27)$$

and

$$b_l^{\mathrm{V2V}} = \begin{cases} 1, \text{ if } \mathrm{P}_r\left(R_l \le R_{\min}^l\right) \le P_{\max}^{\mathrm{outage}}, \forall l, \\ 0, otherwise. \end{cases} \quad (28)$$

In (25), the part 1 signifies the system utility, the part 2 signifies the cost function which is characterized as the satisfactory detection performance, the part 3 and the part 4 denote the cost functions which are represented as the acceptable communication quality constraints of V2I and V2V links, respectively. The parameters $\lambda_1, \lambda_2$ and $\lambda_3$ represent the positive values of the parts 2-4 in (25), respectively, which are employed to trade-off the cost and utility [45]. The purposes of (26), (27) and (28) are to enforce the minimum detection performance and the minimum data rate constraints, respectively. If these requirements are fulfilled in the current time step, that is, $b^{\mathrm{RAD}} = 1$, $b_m^{\mathrm{V2I}} = 1$ or $b_l^{\mathrm{V2V}} = 1$, signifying that there is an encouragement of the reward function owing to the desired detection performance or transmission quality guarantees.

The objective of the training agent is to find an optimal policy $\pi^*$ that optimizes the long-term discounted reward. The cumulative discounted reward is given by

$$R_t = \sum_{i=0}^{N} \gamma^i r_{t+i}, \quad (29)$$

with $\gamma \in (0, 1]$ signifying the discount factor. For a given $\pi$, Q-function with $(s, a)$ at time slot $t$ is denoted by

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{\pi}[R_t | s_t = s, a_t = a]. \quad (30)$$

### B. Hierarchical Integration of DQN and TD3

Considering the fact that the secure performance optimization problem in (22) is a mixed-integer problem with continuous and discrete control variables, we develop a hierarchical learning framework, as presented in Fig. 2, where the DQN and TD3 techniques are employed to enable the agent to train faster in time-varying hybrid RIS-enhanced V2X network. In particular, the binary discrete variable $\rho_{ml}$ decides the spectrum allocation strategy for the V2X links, while the continuous variables

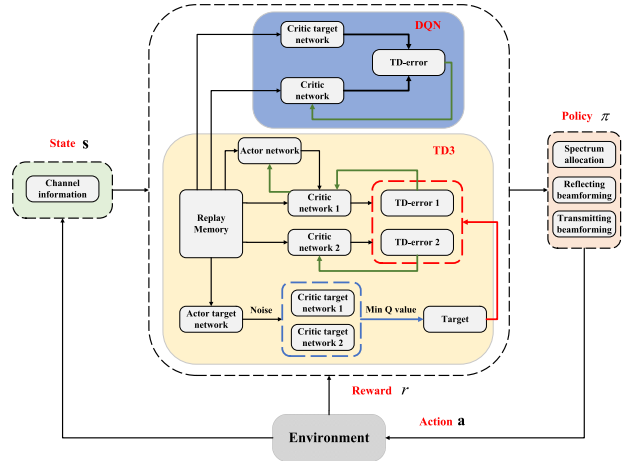


Fig. 2. HTD3-based anti-eavesdropping strategy for hybrid RIS-enhanced V2X network.

contain the RSU's transmit beamforming $\{\mathbf{v}_m\}_{m=1}^M, \mathbf{w}_r$ and the hybrid RIS's passive and active reflection matrices $\mathbf{\Phi}, \mathbf{\Psi}$. The fundamental principle of the hierarchical training scheme is to divide the system action space into discrete part and continuous part, namely $(a^d, a^c)$. The discrete part can be learned employing the traditional DQN algorithm. Specified the discrete part $a^d = \{\rho_{ml}\}_{m\in\mathcal{M}, l\in\mathcal{L}}$ in the outer-loop network architecture, we emphasize on the optimized solutions $a^c = \{\{\mathbf{v}_m\}_{m=1}^M, \mathbf{w}_r, \mathbf{\Phi}, \mathbf{\Psi}\}$ to a continuous control task in (22), which are solved by the inner-loop TD3 procedure. The goal of TD3 technique [35] is to address the function approximation errors that would produce the poor state resulting in suboptimal solutions in DDPG. The HTD3 training scheme permits us to separate the combinatorial optimization of the spectrum allocation strategy from the optimization of continuous secure beamforming variables. The convergence point of the HTD3 procedure is regarded as the Q-value of the DQN algorithm. The advantage of the devised HTD3 scheme primarily lies in the decreased system action space, hence it has the ability to enhance the training capability.

In the proposed hierarchical learning procedure, the outer-loop DQN is used to calculate the discrete Q-function $Q(s, a^d)$ employing the value estimation $Q(s, a^d; \boldsymbol{\theta})$, with $\boldsymbol{\theta}$ being the deep neural network (DNN) weight. The purpose of DQN procedure is to minimize a loss function at current time step, i.e.,

$$\mathcal{L}(\boldsymbol{\theta}_t) = \mathbb{E}\left[\left\{V_t(s_t; \boldsymbol{\theta}_t) - Q(s_t, a_t^d; \boldsymbol{\theta}_t)\right\}^2\right], \quad (31)$$

where $V_t(s_t; \boldsymbol{\theta}_t) = r(s_t, a_t^d) + \gamma \max_{a_{t+1}^d \in \mathcal{A}} Q(s_{t+1}, a_{t+1}^d; \boldsymbol{\theta}_t)$. The target value $V_t(s_t; \boldsymbol{\theta}_t)$ is produced by target Q-network with weights $\boldsymbol{\theta}_t$, which is delayed copy of online Q-network, namely, $\boldsymbol{\theta}_t' = \boldsymbol{\theta}_{t-t_d}$, with $t_d$ being the delay to the current time slot. The error between $V_t(s_t; \boldsymbol{\theta}_t)$ and $Q(s_t, a_t^d; \boldsymbol{\theta}_t)$ is generally referred to as temporal-difference error, i.e.,

$$\delta_t = V_t\left(s_t; \boldsymbol{\theta}_t\right) - Q\left(s_t, a_t^d; \boldsymbol{\theta}_t\right). \quad (32)$$

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The weight $\boldsymbol{\theta}$ can be obtained by taking the partial differentiation of Eq. (31) with respect to $\boldsymbol{\theta}$, that is,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \beta \nabla \mathcal{L}(\boldsymbol{\theta}_t), \tag{33}$$

where $\beta$ denotes the learning rate of the Q-network $\boldsymbol{\theta}$. Therefore, the policy $\pi_t(s)$ of the outer-loop DQN procedure is expressed as

$$\pi_t^{\boldsymbol{\theta}}(s) = \arg\max_{a_t^d \in \mathcal{A}} Q(s_t, a_t^d; \boldsymbol{\theta}_t). \tag{34}$$

The TD3 technique is a modified DDPG procedure, which is also used to deal with continuous action space. We first introduce the DDPG procedure [46]. In the course of the learning, the weights of the DNNs can be trained as follows:

1) *Updating of the Critic Network:* Consider that a mini-batch of $N_B$ transition tuples is sampled from the experience replay buffer $\mathcal{D}$. The target Q-value of the $i$th transition tuple $(s_i, a_i^c, r_i, s_{i+1})$ is expressed as

$$y_i = r_i + \gamma Q\left(s_{i+1}, a_{i+1}^c; \boldsymbol{\theta}^{Q'}\right), \tag{35}$$

where $a_{i+1}^c = \pi(s_{i+1}; \boldsymbol{\theta}^{Q'})$, and $\pi(s_{i+1}; \boldsymbol{\theta}^{Q'})$ signifies a deterministic policy $\pi$ that outputs $a_{i+1}^c$ under $s_{i+1}$ via the target actor network with weights $\boldsymbol{\theta}^{Q'}$. $Q(s_{i+1}, a_{i+1}^c; \boldsymbol{\theta}^{Q'})$ denotes the $Q$-value of continuous action $a_{i+1}^c$ under $s_{i+1}$. By minimizing the mean squared Bellman error between $Q(s_i, a_i^c; \boldsymbol{\theta}^Q)$ with weights $\boldsymbol{\theta}^Q$ and $y_i$, we have

$$\mathcal{L}\left(\boldsymbol{\theta}^Q\right) = \frac{1}{N_B} \sum_{i=1}^{N_B} \left(y_i - Q\left(s_i, a_i^c; \boldsymbol{\theta}^Q\right)\right)^2, \tag{36}$$

with $a_i^c = \pi(s_i; \boldsymbol{\theta}^Q)$. The weights of the critic network $\boldsymbol{\theta}^Q$ is trained through gradient descent, i.e.,

$$\boldsymbol{\theta}^Q = \boldsymbol{\theta}^Q - \beta_Q \nabla_{\boldsymbol{\theta}^Q} \mathcal{L}\left(\boldsymbol{\theta}^Q\right), \tag{37}$$

where $\beta_Q$ indicates the learning rate of $\boldsymbol{\theta}^Q$.

2) *Updating of the Actor Network:* In order to enhance the policy offered by the devised scheme, the weights of the actor network $\boldsymbol{\theta}^\mu$ is trained to let the action output from the actor network be in the direction of increasing the $Q$ value. The weights of the actor network $\boldsymbol{\theta}^\mu$ is trained employing gradient ascent method, i.e.,

$$\boldsymbol{\theta}^\mu = \boldsymbol{\theta}^\mu - \beta_\mu \frac{1}{N_B} \sum_{i=1}^{N_B} \nabla_{a_i^c} Q\left(s_i, a_i^c; \boldsymbol{\theta}^Q\right) \nabla_{\boldsymbol{\theta}^\mu} \pi\left(s_i; \boldsymbol{\theta}^\mu\right), \tag{38}$$

where $a_i^c = \pi(s_i; \boldsymbol{\theta}^\mu)$, $\beta_\mu$ signifies the learning rate of $\boldsymbol{\theta}^\mu$.

3) *Updating of the Target Networks:* The soft update technique [47] is employed to train the weights of the target critic network $\boldsymbol{\theta}^{Q'}$, so that the target value $V(s; \boldsymbol{\theta}^{Q'})$ changes gently in the process of learning, thus ensuring the stability of learning. Therefore, the weight coefficients $\boldsymbol{\theta}^{Q'}$ can be trained using Poylak averaging method with coefficient $\tau$, i.e.,

$$\boldsymbol{\theta}^{Q'} = \tau \boldsymbol{\theta}^Q + (1-\tau)\boldsymbol{\theta}^{Q'}, \quad 0 < \tau \ll 1. \tag{39}$$

Analogously, the weights of the target actor network $\boldsymbol{\theta}^{\mu'}$ can be trained, i.e.,

$$\boldsymbol{\theta}^{\mu'} = \tau \boldsymbol{\theta}^\mu + (1-\tau)\boldsymbol{\theta}^{\mu'}, \quad 0 < \tau \ll 1. \tag{40}$$

To guarantee reasonable exploration of the continuous action spaces, a general exploration strategy is adopted for the agent policy. Specifically, the action is chosen from a random procedure, which is considered as a Gaussian distribution with mean $\mu(s; \boldsymbol{\theta}^\mu)$ and variance $\upsilon\sigma^2$, with $\upsilon$ being a factor to attenuate the randomness of the learning procedure. where $\upsilon$ is a exploration noise and $\sigma$ is a decay constant.

However, DDPG technique has the bias overestimation problem, which would result in an accumulation error. Specifically, the error generates the deteriorated state to admit a large Q value producing a nonoptimal policy. The main objective of inner-loop TD3 technique is to reduce the function approximation errors that produce overestimated value estimates and nonoptimal strategies in DDPG procedure [48]. The inner-loop TD3 framework consists of three networks, containing an actor network and two critic networks. In which, the actor network contains two parts: 1) policy network with DNN weights $\boldsymbol{\theta}^\mu$ and 2) target policy network with DNN weights $\boldsymbol{\theta}^{\mu'}$, and a critic network consists of two parts: 1) main Q network with DNN weights $\boldsymbol{\theta}^Q$ and 2) target Q network with DNN weights $\boldsymbol{\theta}^{Q'}$, which decouple target Q value computation and action selection. TD3 utilizes several supplementary schemes to DDPG procedure to avoid overestimation of the value functions as follows:

1) *Clipped Double Q-learning:* TD3 procedure suppresses persistently the bias overestimation by choosing the smaller target Q value, just as DDQN procedure utilizes double Q-learning technique to address the bias overestimation problem. The TD3 procedure chooses the smaller value function between two actor networks, i.e.,

$$y_i = r\left(s_i, a_i^c\right) + \gamma \min_{j=1,2} \left\{Q'\left(s_{i+1}, \mu'(s_{i+1}) | \boldsymbol{\theta}_j^{Q'}\right)\right\}. \tag{41}$$

2) *Delayed Policy Updates:* The scheme gradually trains policy network after training the critic network precisely in preparation for inaccurate value functions if policy training can not work validly. When the critical network is trained twice, the policy and target networks need to be learned once. Furthermore, TD3 utilizes delayed policy updates to reduce the growth of error.

3) *Target Policy Smoothing:* DDPG procedure may generate an overestimation of the value function of the policy when a peak appears in achieving the policy from the actor network. A smoothing regularization scheme for the Q-function can prevent the policy network from overfitting. TD3 procedure is capable of ameliorating the issue by appropriately smoothing the target policy via the clipped random noise addition to the action, that is,

$$a_{i+1}^c = \mu'(s_{i+1}) + [\varepsilon]_{-c}^c, \tag{42}$$

where $\varepsilon$ represents a Gaussian random variable with distribution $\varepsilon \sim \mathcal{CN}(0, \sigma_s^2)$ and $c$ denotes a noise constant.

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The clipped random noise is viewed as a regularization method that smoothes the Q-values within a limited range $[-c, c]$ around the action $\mu'(s_{i+1})$.

Based on the above definitions and discussions, the proposed HTD3-based secure communication performance optimization algorithm for the hybrid RIS-enhanced V2X network is summarized in Algorithm 1. The central control unit at the RSU gathers V2X network information and optimizes secure beamforming and spectrum resource allocation. During the learning phase, the central control unit initializes all neural network weights as well as gathers the current state containing the CSI of all vehicles, previous radar MI, secrecy rate of all CVUs, data rate of all CVUs and DVUs. The system state is input into the HTD3 framework to train the network parameters. Specified the state, the outer-loop DQN procedure determines the binary spectrum allocation indicator $\rho_{ml}$ for each V2X link, and then the channel coefficients are updated based on the spectrum sharing protocol. The discrete action $a^d$ is selected by using the probability distribution $\pi^{\boldsymbol{\theta}}(s, a^d)$. By maximizing the secure communication performance of all V2I links, the neural networks of the inner TD3 produce the continuous beamforming variables and the value estimate. Specifically, in each time slot, the proposed inner-loop TD3 procedure estimates continuous action $a^c$ from the current actor network $\mu(s; \boldsymbol{\theta}^{\mu})$, adds the noise and computes the immediate reward. HTD3 procedure updates channel information at each time step and next system state accordingly. A replay buffer with the size $\mathcal{D}$ is utilized for learning the actor and critic networks, and initialized at the beginning. To be precise, HTD3 stores the environmental model data into the replay buffer, and extracts a small batch of data from the buffer to learn the parameters of the actor networks and critic networks.

The learning phase requires a high performance computing server which is implemented offline at the RSU while the execution phase is accomplished online. The neural network parameters need to be trained only when the external environment has experienced significantly varies, primarily relying on the channel dynamics and communication and sensing requirements. After effective learning in Algorithm 1, the HTD3 framework is loaded for the execution phase. In the execution phase, the central control unit employs the learned HTD3 framework to output its chosen action using the network parameters, with the obtained state from the hybrid RIS-enhanced secure V2X network. Then, the considered V2X network feeds back an immediate reward value and a current state to the hierarchical learning agent. Finally, the RSU's transmit beamforming vectors, the hybrid RIS's active and passive reflection weights and spectrum allocation strategy are attained based on the selected action.

**Algorithm 1:** HTD3 Algorithm for Joint Secure Beamforming Optimization and Spectrum Resource Allocation.

**Input:** $\mathbf{h}_m$, $\mathbf{h}_l$, $\mathbf{h}_E$, $P^{\mathrm{R}}_{\max}$, $P^{\mathrm{RIS}}_{\max}$, minimum data rate requirements of all V2X links, outage probability of V2V links, minimum radar MI threshold.
**Outout:** optimal action $a = \{\{\mathbf{v}_m\}_{m=1}^{M}, \mathbf{w}_r, \boldsymbol{\Phi}, \boldsymbol{\Psi}, \boldsymbol{\rho}\}$.
**Initialize:** experience replay buffer $\mathcal{D}$, online Q-network, critic networks, and actor network with random parameters $\boldsymbol{\theta}$, $\boldsymbol{\theta}_1^Q$, $\boldsymbol{\theta}_2^Q$, $\boldsymbol{\theta}^{\mu}$, target networks $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}$, $\boldsymbol{\theta}_1^{Q'} \leftarrow \boldsymbol{\theta}_1^Q$, $\boldsymbol{\theta}_2^{Q'} \leftarrow \boldsymbol{\theta}_2^Q$, $\boldsymbol{\theta}^{\mu'} \leftarrow \boldsymbol{\theta}^{\mu}$, spectrum allocation variables $\boldsymbol{\rho}$, beamforming vectors $\{\mathbf{v}_m\}_{m=1}^{M}$ and $\mathbf{w}_r$, active RIS matrix $\boldsymbol{\Phi}$, passive RIS matrix $\boldsymbol{\Psi}$.

**for** each episode **do**
  **for** each time step $t$ **do**
    Observe system state $s_t$;
    Select discrete action $a_t^d$ based on $\epsilon$-greedy policy;
    Select continuous action $a_t^c$ using (42);
    Observe a reward $r_t$;
    Update vehicle/eavesdropper locations and the corresponding path loss coefficients;
    Observe $s_{t+1}$;
    Store $(s_t, (a_t^d, a_t^c), r_t, s_{t+1})$ in the replay buffer $\mathcal{D}$;
  **end for**
  **if** $R > 2000$ **then**
    Sample mini-batch of $N_B$ transitions $(s_i, (a_i^d, a_i^c), r_i, s_{i+1})$ from $\mathcal{D}$;
    Obtain loss function for critic networks using (41);
    Update critic network weights $\boldsymbol{\theta}_j^Q, j = 1, 2$ using (37);
    Update Q-network weights $\boldsymbol{\theta}$ using (33);
    **if** $t$ mod $d$ **then**
      Update actor network weights $\boldsymbol{\theta}^{\mu}$ using (38);
      Update target critic network weights $\boldsymbol{\theta}_j^{Q'}, j = 1, 2$ using (39);
      Update target actor network weights $\boldsymbol{\theta}^{\mu'}$ using (40);
      Update target Q-network weights: $\boldsymbol{\theta}_t' \leftarrow \boldsymbol{\theta}_{t-t_d}$;
    **end if**
  **end if**
**end for**

### C. Computational Complexity

To assess the computational complexity of the proposed hierarchical algorithm, the floating-point operations per second (FPOS) is employed as the metric in the training phase. For any fully connected layer $\mathrm{L}_i$ of input size $\mathrm{I}_i$ (which is proportional to the number of system state) and output size $\mathrm{O}_i$, the number of FPOS is expressed by $\mathrm{FPOS}(\mathrm{L}_i) = 2\mathrm{I}_i\mathrm{O}_i$. Therefore, the total number of FPOS for actor network with L hidden layers yields:

$$\sum_{i=1}^{3} \mathrm{FPOS}_{\mathrm{Actor}}(\mathrm{L}_i) = 2\left(H_1 |\mathcal{S}| + H_L |\mathcal{A}| + \sum_{i=2}^{L} H_{i-1}H_i\right)$$
$$= 2H_1 (M(N_t + N_r + 1) + N_r) + 2H_L (N_h + N_v + M)$$
$$+ 2\sum_{i=2}^{L} H_{i-1}H_i, \tag{43}$$

where $H_1$, $H_L$ and $H_i$ denote the size of the first hidden layer, $L$-th hidden layer, and $i$-th hidden layer, respectively. Analogously, the total number of FPOS for critic network with L hidden

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layers yields:

$$\sum_{i=1}^{3} \mathrm{FPOS}_{\mathrm{Critic}}\left(\mathrm{L}_i\right) = 2\left(H_1\left|\mathcal{S}\right| + H_L + \sum_{i=2}^{L} H_{i-1}H_i\right)$$
$$= 2\left(H_1\left(M\left(N_t + N_r + 1\right) + N_r\right) + H_L + \sum_{i=2}^{L} H_{i-1}H_i\right). \tag{44}$$

The policy implemented in this paper has 2 hidden layers of sizes 258 and 128. The number of FPOS during the inference is mainly determined by the matrix multiplications of the policy network, which has four layers with size $|\mathcal{S}|$, 256, 128, and $|\mathcal{A}|$. The amount of FPOS for a connected layer is twice the product of the size of the hidden layers. That is, specified a hidden layer of size 128 that has a previous layer of size 256, the weight matrix should be of size $128 \times 256$ and the FPOS is $2 \times (128 \times 256)$. According to the aforementioned analysis, the computational complexity of the proposed HTD3 algorithm is

$$2\left(H_1^{\mathrm{Critic}}\left|\mathcal{S}\right| + H_L^{\mathrm{Critic}} + \sum_{i=2}^{L} H_{i-1}^{\mathrm{Critic}} H_i^{\mathrm{Critic}}\right)$$
$$+ 2\left(H_1^{\mathrm{Actor}}\left|\mathcal{S}\right| + H_L^{\mathrm{Actor}}\left|\mathcal{A}^c\right| + \sum_{i=2}^{L} H_{i-1}^{\mathrm{Actor}} H_i^{\mathrm{Actor}}\right)$$
$$+ 2\left(H_1^{\mathrm{DQN}}\left|\mathcal{S}\right| + H_L^{\mathrm{DQN}}\left|\mathcal{A}^d\right| + \sum_{i=2}^{L} H_{i-1}^{\mathrm{DQN}} H_i^{\mathrm{DQN}}\right).$$

It shows that the complexity of the devised hierarchical procedure is higher than the traditional DQN learning technique. Nevertheless, our hierarchical procedure realizes superior performance than that of the traditional DQN technique, which will be presented in the next section. Note that the high computational complexity of the DNN learning stage can be implemented offline for a finite number of episodes at the central control unit.

## IV. Numerical Results

### A. Parameter Settings

This section offers simulation experiments to confirm the devised HTD3-based secure communication scheme for the hybrid RIS-enhanced V2X network. Unless otherwise stated, the RSU is designed with an uniform linear array (ULA) of $N_t = 16$ transmit and $N_r = 8$ receive antennas. We assume that the power budget of RSU $P_{\max}^{\mathrm{R}}$ ranges from 16 dBm to 35 dBm, the transmit power of DVU $P_{\max}^{\mathrm{D}}$ varies from 16 dBm to 25 dBm, the number of hybrid RIS units $N_{\mathrm{I}}$ ranges from 30 to 60, the outdated CSI factor $\rho_d$ ranges between 0.6 and 1 for various simulation settings. We set the path loss exponent of the RSU-CVU link as $\alpha_{\mathrm{R}m} = 3$, since this channel has plenty of obstacles and scatterers [15]. For the channel between the RSU and the RIS and that between the RIS and the CVU, LoS components exist in practical implementation. The parameter $\alpha_{\mathrm{RI}} = \alpha_{\mathrm{I}m} = 2$ is used to signify the path loss exponent of the RSU-RIS and RIS-CVU links. Similarly, the path loss exponents of the RSU-Eve and RIS-Eve links are assumed to be $\alpha_{\mathrm{RE}} = 3$ and $\alpha_{\mathrm{IE}} = 2$, respectively. Rician fading factor $\kappa = 3$dB for all the LoS cases is considered.

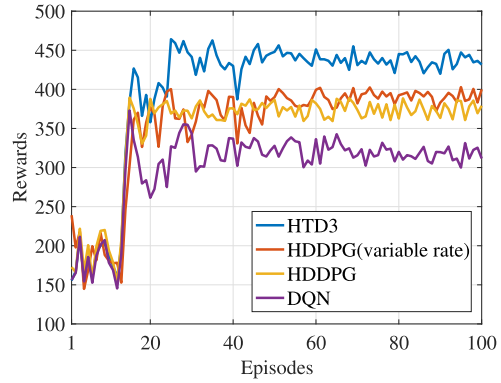


Fig. 3. Convergence property of different DRL-based techniques versus episodes.

Regarding the proposed hierarchical learning framework, the DQN contains 2-fully connected hidden layers with neurons 128 and 64, respectively. Adam optimizer is employed to train neural network weights. As for TD3, the actor networks and the critic networks are designed by using 2-layer networks with sizes 256 and 128, respectively. For the actor network, $\tanh(\cdot)$ is considered as the activation function to bound the actions. Main system simulation parameters are given in Table I. Furthermore, we compare our HTD3-based joint beamforming optimization and spectrum allocation technique with the following methods:

- The hierarchical DDPG (HDDPG) [49] and variable rate HDDPG [50] approaches, which are used to design the secure beamforming and allocate the spectrum in RIS-enhanced vehicular networks.
- The improved DQN technique [31], where DNN is utilized to calculate the Q-value function, when selecting the secure communication strategy corresponding to the greatest Q-value.
- The scheme adopts successive convex approximation (SCA) technique to tackle the secure communication problem (22), and alternate optimization (AO) procedure is employed to design transmit beamforming and allocate spectrum resource [15].
- The transmit beamforming of RSU and the reflection weights of hybrid RIS are designed without spectrum sharing scheme (beamforming design only).

### B. Convergence

First, the convergence property of various DRL-based approaches is compared, considering $N_{\mathrm{I}} = 30$, $P_{\max}^{\mathrm{R}} = 25$ dBm and $P_{\max}^{\mathrm{D}} = 18$ dBm. The reward with increasing learning episodes to explore the convergence performance of the devised scheme and three compared techniques is presented in Fig. 3. Obviously, our hierarchical training scheme realizes the better performance than that of the HDDPG [49] and self-adjusting HDDPG [50] techniques. Particularly, the proposed HTD3-based scheme depends on the double Q-learning and smoothing regularization techniques to reduce the overestimation bias and

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TABLE I
SIMULATION PARAMETERS

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| Number of CVUs $M$ | 4 | Number of DVUs $L$ | 3 |
| Target detection coefficient $\alpha_{\mathrm{E}}$ | 0.5 | Maximum power of active RIS $P_{\max}^{\mathrm{RIS}}$ | 20 dBm |
| Minimum data rate of V2I link $R_{\min}^{m}$ | 4 bps/Hz | Minimum data rate of V2V link $R_{\min}^{l}$ | 5 bps/Hz |
| The V2V outage probability $P_{\max}^{\mathrm{outage}}$ | 0.9 | Radar MI threshold $\mathrm{MI}_{\min}$ | 0.3 bit |
| Cost parameters $\lambda_1$, $\lambda_2$, $\lambda_3$ | 0.1, 0.02, 0.02 | Experience replay buffer capacity $\mathcal{D}$ | 500 |
| Gaussian exploration noise $\varepsilon$ | $\mathcal{CN}(0, 0.02)$ | Mini-batch size $N_B$ | 64 |
| Target network Polyak averaging rate $\tau$ | 0.02 | Learning rate for Actor and critic networks $\beta_Q$, $\beta_\mu$ | 0.002, 0.001 |
| Discounted rate $\gamma$ | 0.9 | Learning rate for Q-network $\beta$ | 0.001 |

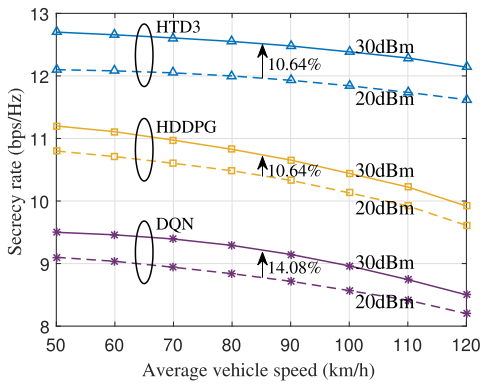


Fig. 4. Sum secrecy rate of all V2I links with varying average vehicle speed.

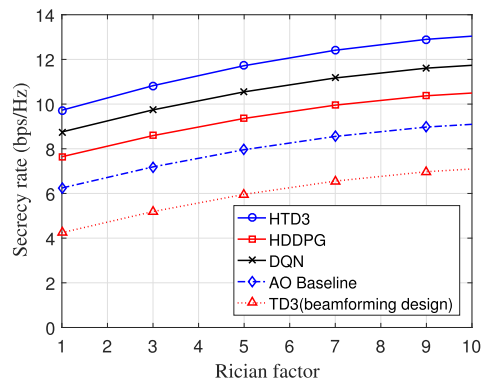


Fig. 5. Sum secrecy rate of all V2I links with varying Rician factor.

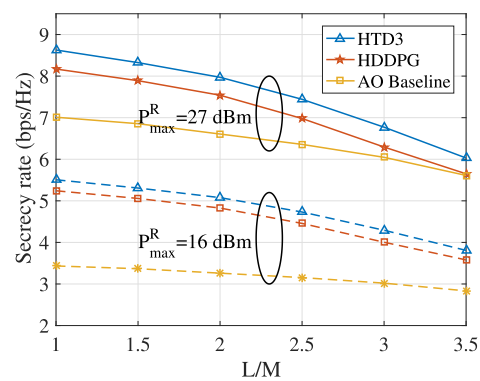


Fig. 6. Sum V2I secrecy rate with different number of active V2V links.

improve the network stable, and the enhanced policy is obtained after a certain amount of learning episodes. The self-adjusting HDDPG approach is able to attain the higher convergence rate by exploiting variable rate learning strategy, but it obtains inferior performance than that of the HTD3 scheme, producing the smaller reward value. Furthermore, as expected, even though the DQN-based training [31] scheme has the relatively simple architecture, the worst capability fails to improve the system performance with improving learning episodes.

As presented in Fig. 3, the reward is enhanced as the number of iterations increases, verifying the usefulness of the devised hierarchical training method. When the iteration number reaches about 45, the reward progressively converges in spite of small fluctuations owing to channel variation as well as the $\varepsilon$-greedy strategy in the learning procedure. Despite the hybrid RIS-enhanced V2X network is dynamic and the system state is large, the proposed scheme only needs approximately 50 iterations to appropriately converge.

We study the sum secrecy rate of all V2I links against the average vehicle speed in Fig. 4. As expected, the sum V2I secrecy rate attained by all approaches decreases as the average speed increases, and the secrecy performance of decrease of the proposed strategy is lower than that of the counterparts. Moreover, when the power budget of RSU increases, the V2I secrecy rate increases accordingly. The results are ascribed to the dominant reason. A higher speed leads to sparser traffic based on the considered vehicular network setup, which increases the distance between vehicles. To ensure the reliability of the active V2V links, DVUs will increase the power budget to offset for the higher path loss induced by the increased distance between vehicles, which would produce stronger disturbance to the CVUs and hence limits the secrecy performance improvement of V2I links. The effect of various Rician factors on the secure performance is depicted in Fig. 5. The sum V2I secrecy rate of all approaches enhances as the value of $\kappa$ increases. The result was expected since all links tend to be more deterministic as Rician factor increases, and a larger proportion of LoS components is achieved to improve the secure communication performance.

To evaluate the effectiveness of spectrum reuse scheme on the secure communication performance for the hybrid RIS-enhanced V2X network, the requirement $\sum_{m\in M}\rho_{ml}\leq 1, \forall l$ is ignored, which implies that the spectrum utilized by each V2I link is coexisted by multiple active V2V links. From Fig. 6, the sum V2I secrecy rate generated by three different methods

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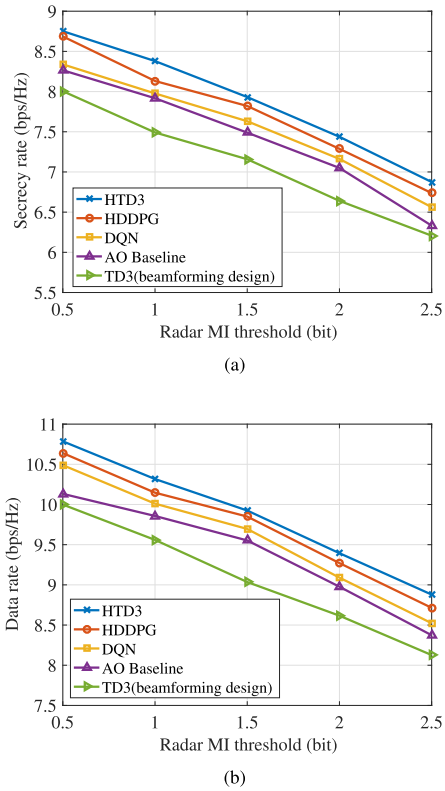


Fig. 7. Performance comparisons for different radar MI threshold. (a) communication secrecy rate, (b) communication data rate.

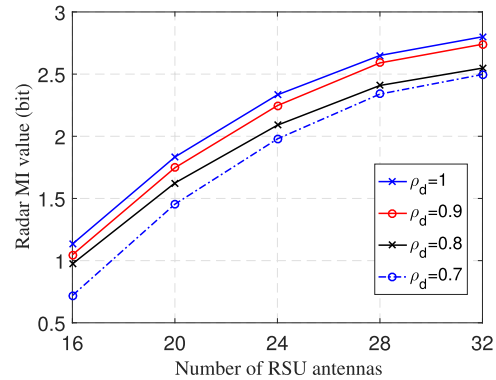


Fig. 8. Radar MI value versus the number of RSU antennas.

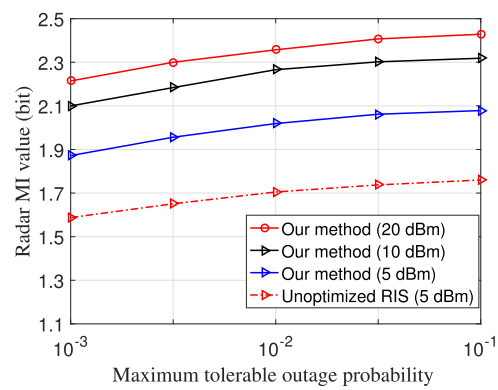


Fig. 9. Radar MI versus different maximum tolerable outage probability.

reduces as the number of spectrum sharing active V2V links increases. The principal reasons for secure performance deterioration are as follows. All V2I links are required to share channel with several V2V links at the same time. To ensure the validity of active V2V links, the power budget of the CVUs should be regulated to minimize the disturbance to active V2V links. The reduced power budget of all V2I links results in the degeneration of the sum V2I secrecy rate. Moreover, an increase in number of active V2V links will cause more disturbance to the available V2I links and thus decrease $\mathrm{SINR}_m^{\mathrm{C}}$ value. When the power budget of RSU is increased from 16 dBm to 27 dBm, the V2I secrecy performance increases accordingly. If the condition is satisfied $L/M \geq 3$, the secure performance improvement is marginal. We can observe that in this case, the available V2I links suffer serious disturbance from active V2V links, which is the principal reason that adversely influences the secure communication.

The sum secrecy rate and data rate of all V2I links against the radar MI level for $P_{\max}^{\mathrm{R}} = 25\,\mathrm{dBm}$ and $P_{\max}^{\mathrm{D}} = 18\,\mathrm{dBm}$, are shown in Fig. 7 for five different approaches. From Fig. 7, both the sum secrecy rate and data rate decrease as the radar MI threshold increases for all techniques. That is because as the radar detection performance demand increases, the hybrid RIS-enhanced RSU takes more energy to focus the beamformer of ISAC signals at the direction of the malicious target, and hence less power is utilized for the communication waveform transmission toward the direction of the connected CVUs. The devised HTD3 technique provides higher sum secrecy rate and data rate for various radar MI values as compared to the counterparts.

We explore the influence of number of RSU antennas for radar MI performance under different outdated CSI factor with $P_{\max}^{\mathrm{RIS}} = 20\,\mathrm{dBm}$ and $N_I = 60$. From Fig. 8, the radar MI value increases as the increase of the number of RSU antennas. The phenomenon is owing to the additional spatial degree of freedoms (DoFs) introduced by the more RSU antennas, and then improve the beamformer resolution. Moreover, the RIS-related CSI error influences the sensing capability from two aspects: on the one hand, each DVU requires more energy to satisfy the data rate constraints for a higher CSI error level; on the other hand, the MI degeneration is caused by the sensing part to address worse RSU-RIS channel condition. Fig. 9 illustrates that the radar MI value increases with the maximum tolerable outage probability $P_{\max}^{\mathrm{outage}}$ increases. The power budget of the hybrid RIS is assumed as $P_{\max}^{\mathrm{RIS}} \in \{5\,\mathrm{dBm}, 10\,\mathrm{dBm}, 20\,\mathrm{dBm}\}$. With the increase of $P_{\max}^{\mathrm{outage}}$, more relaxed constraint is imposed on the data rate of V2V links, and less power is required on the communication part to satisfy the transmission rate constrained, hence improve the sensing performance. Furthermore, the sensing performance improves with increasing the power consumption of the hybrid RIS $P_{\max}^{\mathrm{RIS}}$.

Fig. 10 illustrates the radar MI level versus the number of active units on the hybrid RIS $A$ of the proposed method for $P_{\max}^{\mathrm{RIS}} \in \{5\,\mathrm{dBm}, 10\,\mathrm{dBm}, 20\,\mathrm{dBm}\}$, $N_I = 60$ and optimized passive RIS baseline is also considered as a benchmark. When

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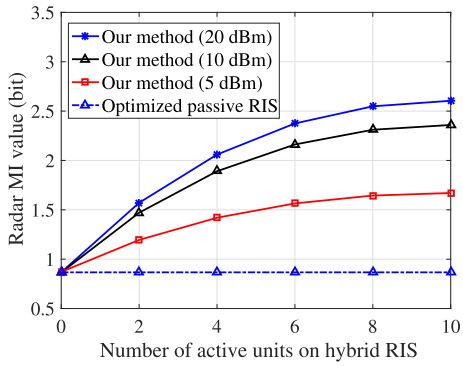


Fig. 10. Radar MI value versus the number of active units on hybrid RIS for $P_{\max}^{\mathrm{RIS}} \in \{5\,\mathrm{dBm}, 10\,\mathrm{dBm}, 20\,\mathrm{dBm}\}$.

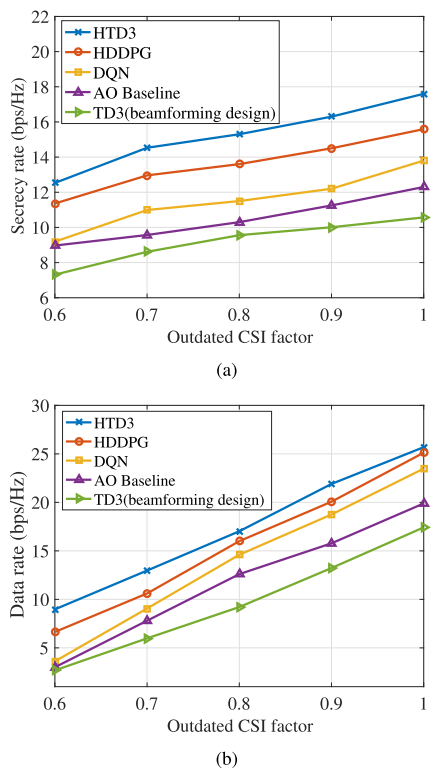


Fig. 11. Performance comparisons for different outdated CSI factor. (a) communication secrecy rate, (b) communication data rate.

$A = 0$, the proposed approach is reduced to the passive RIS, thus they have the similar radar MI performance. In addition, the radar MI performance of our approach improves as the value of $A$ increases. The result indicates that the hybrid RIS strategy dramatically enhances the radar sensing performance by introducing a small amount of active elements to conventional passive RIS.

Then, we evaluate how the sum secrecy rate and data rate are influenced by the outdated channel factor $\rho_d$ in the considered V2X network, that is, from $\rho_d = 0.5$ to 1, when $N_{\mathrm{I}} = 30$, $P_{\max}^{\mathrm{R}} = 25\,\mathrm{dBm}$, and $P_{\max}^{\mathrm{D}} = 18\,\mathrm{dBm}$. With the decrease of $\rho_d$ value, the transmission channel gets more outdated as described in (5), and $\rho_d = 1$ indicates perfect CSI. From Fig. 11, when channel gets more outdated (as the value of $\rho_d$ reduces), the sum secrecy rate and data rate decrease for all optimization methods. This is because a greater $\rho_d$ value signifies more precise CSI, which would enable all methods to optimize spectrum allocation and secure beamforming to realize higher sum secrecy rate and data rate of all V2I links. We can observe that decreasing $\rho_d$ value has more influences on the performance gain improvement of the counterparts while the HTD3 technique still keeps the communication security and quality at satisfactory level, suggesting that the counterparts are less robust to the fluctuation of communication channel and the robust of the devised technique. In particular, our scheme attains the sum secrecy rate and data rate enhancements of 14.28% and 12.41%, compared with the HDDPG method when $\rho_d = 0.9$. Furthermore, in comparison, our method realizes the best communication security and quality among all methods against environment uncertainty. This is due to the fact that the devised method takes into account the time variant channels and employs TD3-tranining to validly learn the dynamic V2X network.

Fig. 12 illustrates the system secrecy rate, data rate and radar MI versus the power budget of RSU $P_{\max}^{\mathrm{R}}$, considering $P_{\max}^{\mathrm{D}} = 18\,\mathrm{dBm}$, $N_{\mathrm{I}} = 30$ and $\rho = 0.7$. From Figs. 12(a) and (b), the system secrecy rate and data rate of five different methods increase with increasing $P_{\max}^{\mathrm{R}}$. As expected, when $P_{\max}^{\mathrm{R}}$ increases, the SINR at each CVU increases, resulting in the communication performance enhancement. Furthermore, as we can see, our training scheme outperforms the AO method. Indeed, our learning strategy optimizes the secure beamforming and allocates the spectrum resource, which synchronously facilitates higher channel transmission benefit for all CVUs and disturb Eve, while the AO baseline method designs the secure beamforming in an alternating optimization manner. The HTD3-based technique has superior performance than HDDPG and DQN in terms of system secrecy rate and data rate. This is due to the fact that delayed policy update scheme is used to reduce the function approximation errors and thus ensure efficient learning capacity. As expected, the proposed hierarchical learning scheme offers noteworthy higher secrecy rate and data rate than the hybrid RIS-enhanced V2X network without spectrum allocation strategy. At the same time, the radar MI level versus $P_{\max}^{\mathrm{R}}$ for various methods is displayed in Fig. 12(c), which demonstrates that the target detection performance becomes more excellent, as $P_{\max}^{\mathrm{R}}$ increases. The results show that our scheme can effectively ensure secure communication and radar sensing requirement via joint beamforming optimization and spectrum allocation.

The sum secrecy rate, data rate and radar MI of all techniques are investigated varying the reflection elements, namely, from $N_{\mathrm{I}} = 30$ to 60, considering $\rho_d = 0.7$, $P_{\max}^{\mathrm{R}} = 25\,\mathrm{dBm}$ and $P_{\max}^{\mathrm{D}} = 20\,\mathrm{dBm}$. As shown in Fig. 13(a) and (b), the sum secrecy rate and data rate of all hybrid RIS-enhanced optimization approaches prominently increase with the value of $N_{\mathrm{I}}$. As expected, more hybrid reflection units, more transmission power and propagation paths are reflected by the hybrid RIS to enhance the received SINR at the CVUs but to deteriorate the received SINR at the Eve. Moreover, the sum secrecy rate of the devised training method is higher than that of the AO and DQN methods, particularly, the performance gap evidently increases with $N_{\mathrm{I}}$, this is because that with more hybrid RIS units, the HTD3-based

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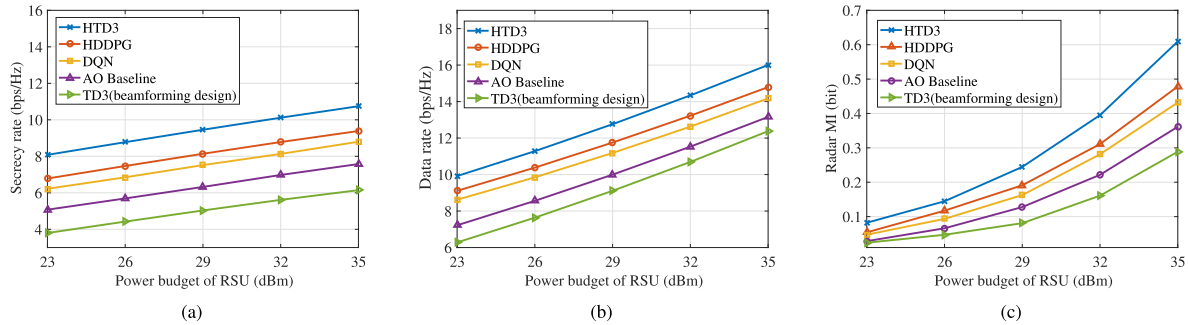


Fig. 12. Performance comparisons versus the power budget of RSU. (a) communication secrecy rate, (b) communication data rate, (c) radar MI level.

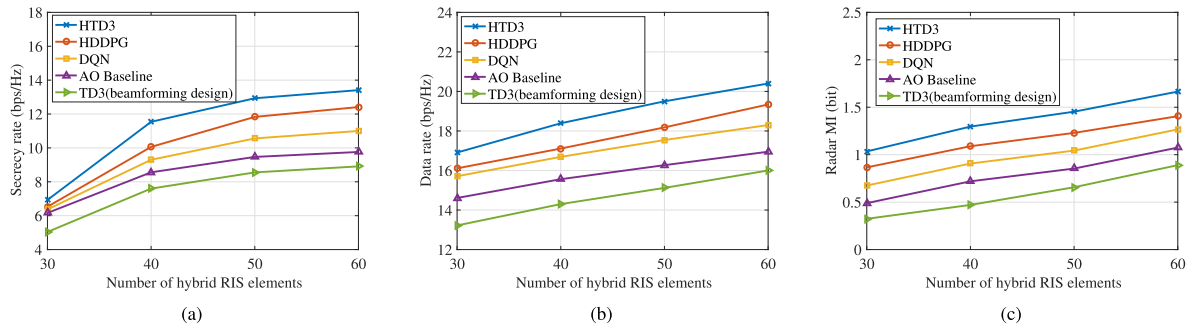


Fig. 13. Performance comparisons versus the number of hybrid reflection elements. (a) communication secrecy rate, (b) communication data rate, (c) radar MI level.

secure transmission technique gets more flexible for reflection beamforming optimization and thus realizes greater performance improvements. Similarly, from Fig. 13(c), the radar MI level increases with the increasing of $N_{\mathrm{I}}$ as more hybrid RIS units are capable of offering reflection beamforming gain to improve target detection capability. Specifically, compared to the hybrid RIS-enhanced method with 30-element, it can be found that a 60-element hybrid RIS provides about 0.8 b radar MI gain, since the employment of hybrid RIS produces more signal paths and enables extra DoFs for enhancing the radar detection capability.

Finally, to further demonstrate the advantages of the MI metric, Fig. 14(a) illustrates the root mean square error (RMSE) of target angle estimation for various radar performance metrics versus the radar MI which is defined in (17). Here, we consider $R^{m}_{\min} = 5$ bps/Hz for all V2I links, and employ the maximum likelihood estimation (MLE) to estimate the angle of the malicious eavesdropping target. We can observe that when radar MI level is high, the RMSE of the proposed scheme is better than that of the techniques based on the beampattern metric [2] and the MSE metric [51], but slightly worse than that of the scheme based on the Cramer-Rao bound (CRB) metric [42]. On the contrary, at low radar MI level (below 0.9 dB), the RMSE of the method based on the MI metric outperforms the other three benchmark techniques. The result further confirms that the ISAC secure scheme based on the MI metric can obtain superior radar sensing capability. Fig. 14(b) shows the sensing RMSE versus the radar MI level. We can observe from the curves that the RMSE of the proposed scheme is reduced with the increase of radar MI value. On the other hand, when we gradually reduce the data rate threshold of the V2I link from

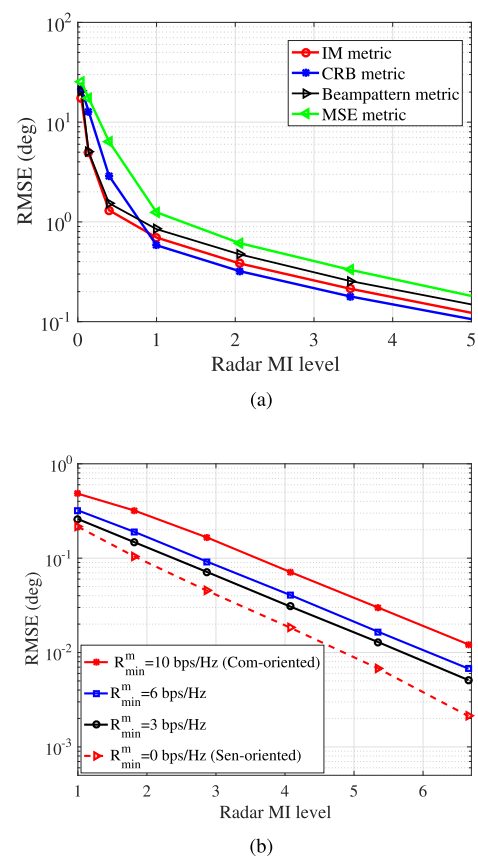


Fig. 14. RMSE of target angle estimation versus the radar MI level. (a) various radar performance metrics for $R^{m}_{\min} = 5$ bps/Hz, (b) various transmission rate threshold $R^{m}_{\min}$.

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10 bps/Hz to 0 bps/Hz, corresponding to the four curves in the figures, the considered system is accordingly converted from communication-dominated to sensing-dominated. As a result, it is observed that the radar sensing performance becomes better gradually.

## V. Conclusion

This paper proposed an optimization technique for securing the hybrid active-passive RIS-aided V2X network, which involved the joint design of the transmit beamforming of RSU, the spectrum reuse scheme of V2X links, the active and passive reflection weights of hybrid RIS. The joint design was formulated as an optimization problem to maximize the sum V2I secrecy capacity subject to radar sensing performance and overall power budget constraints. Since the system model is dynamic, an efficient HTD3 learning algorithm has been developed, where DQN technique has been integrated into the TD3 procedure in a hierarchical framework for obtaining the optimal anti-eavesdropping policy. The simulation results have confirmed the noteworthy improvement of the sum V2I secrecy capacity while ensuring the V2V communication quality requirement and desired RSU target detection performance under a particular resource constraint.

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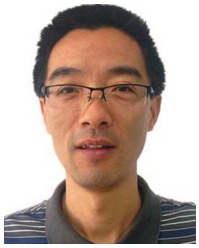

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